1) PROGRAM:

```
# Run theHarvester against the authorised lab domain (two sources)
theHarvester -d testlab.local -b bing -f bing.xml
theHarvester -d testlab.local -b duckduckgo -f ddg.xml

# De-duplicate and sort the discovered hosts/emails
#!/bin/bash

grep -rhoE '[a-zA-Z0-9._-]+@testlab\.local' bing.* ddg.* | sort -u > emails.txt

grep -rhoE '([a-z0-9-]+\.)+testlab\.local' bing.* ddg.* | sort -u > hosts.txt

{
  echo "ASSET,SOURCE"
  sed 's/$/,theHarvester/' emails.txt
  sed 's/$/,theHarvester/' hosts.txt
} > osint_inventory.csv

cat osint_inventory.csv
```

OUTPUT:

2)PROGRAM:

```
 #!/bin/bash
# discover.sh <CIDR> — authorised subnet only (simplified /24)

cidr="$1"
[ -z "$cidr" ] && { echo "Usage: $0 <CIDR>"; exit 1; }

net=(echo"cidr" | cut -d/ -f1 | cut -d. -f1-3)

echo "ip,live,open_ports,flagged" > inventory.csv

scanned=0
live=0
flagged=0

for i in $(seq 1 254); do
  ip="net.i"
  scanned=$((scanned+1))

  if ping -c1 -W1 "$ip" &>/dev/null; then
```

```
    live=$((live+1))

    ports=(nmap − T4 −− open − p1 − 1024"ip" 2>/dev/null |
      awk -F/ '/open/{printf "%s ",$1}')

    flag="OK"

    echo "$ports" | grep -qE '\b(21|23|445)\b' && {
      flag="REVIEW"
      flagged=$((flagged+1))
    }

    echo "$ip,yes,\"$ports\",$flag" >> inventory.csv
  fi
done

echo "Hosts scanned: $scanned, live: $live, flagged: $flagged"
```

OUTPUT:

```
$ sudo ./discover.sh 192.168.56.0/24
Hosts scanned: 254, live: 3, flagged: 2

$ cat inventory.csv
ip,live,open_ports,flagged
192.168.56.102,yes,"21 22 80 445 3306 ",REVIEW
192.168.56.105,yes,"22 23 80 ",REVIEW
192.168.56.108,yes,"22 443 ",OK
```

3)PROGRAM:

```
#!/bin/bash
# webcheck.sh <base_url> — authorised staging host only (GET/HEAD)

url="$1"
[ -z "$url" ] && { echo "Usage: $0 <base_url>"; exit 1; }

host=(echo"url" | awk -F/ '{print $3}')
rep="webreport_${host}.txt"
: > "$rep"

hdr=(curl − s − I"url")

echo "hdr"|grep − i"Server: "|tee − a"rep"

miss=0
```

```
for h in Strict-Transport-Security Content-Security-Policy X-Frame-Options
X-Content-Type-Options; do
  echo "hdr"|grep − qi"h" || {
    echo "MISSING: h"|tee − a"rep"
    miss=$((miss+1))
  }
done

exp=0

for p in /.git/ /backup.zip /admin/ /config.php.bak /phpinfo.php; do
  code=(curl − s − o/dev/null − w"%http_code""url$p")

  [ "$code" = "200" ] && {
    echo "EXPOSED: p"|tee − a"rep"
    exp=$((exp+1))
  }
done

curl -s "$url/" | grep -qi "Index of" &&
  echo "Directory listing enabled" | tee -a "$rep"

command -v nikto >/dev/null &&
  nikto -h "url"2 > /dev/null|tail − 5 >> "rep"

echo "Missing headers: $miss, Exposed paths: $exp"
```

OUTPUT:

```
$ ./webcheck.sh http://192.168.56.102/dvwa

Server: Apache/2.2.8 (Ubuntu) PHP/5.2.4
MISSING: Strict-Transport-Security
MISSING: Content-Security-Policy
MISSING: X-Frame-Options
EXPOSED: /config.php.bak
Directory listing enabled
Missing headers: 3, Exposed paths: 1
```

4)PROGRAM:

```
#!/bin/bash
# audit.sh — local READ-ONLY hardening audit (modifies nothing)

rep="hardening_report.txt"
: > "$rep"
```

```
high=0

echo "== World-writable (first 20) ==" | tee -a "$rep"
find /etc /var "$HOME" -xdev -perm -0002 2>/dev/null |
  head -20 | tee -a "$rep"

echo "== SUID/SGID binaries ==" | tee -a "$rep"
find / -xdev − perm − 4000 − o − perm − 2000 2>/dev/null |
  head -20 | tee -a "$rep"

for f in /etc/hosts.equiv "$HOME/.rhosts"; do
  [ -f "$f" ] && {
    echo "HIGH RISK: fpresent"|tee − a"rep"
    high=$((high+1))
  }
done

if grep -qE '^\s*[^#].*\*\s*\(' /etc/exports 2>/dev/null; then
  echo "HIGH RISK: NFS export to * in /etc/exports" | tee -a "$rep"
  high=$((high+1))
fi

echo "== Listening services ==" | tee -a "$rep"
ss -tlnp 2>/dev/null | tee -a "$rep"

echo "HIGH RISK findings: $high"
```

OUTPUT:

```
$ ./audit.sh
== World-writable (first 20) ==
/var/tmp/shared

== SUID/SGID binaries ==
/usr/bin/passwd
/usr/bin/find

HIGH RISK: /etc/hosts.equiv present
HIGH RISK: NFS export to * in /etc/exports

== Listening services ==
LISTEN 0 128 0.0.0.0:22
LISTEN 0 128 0.0.0.0:3306

HIGH RISK findings: 2
```